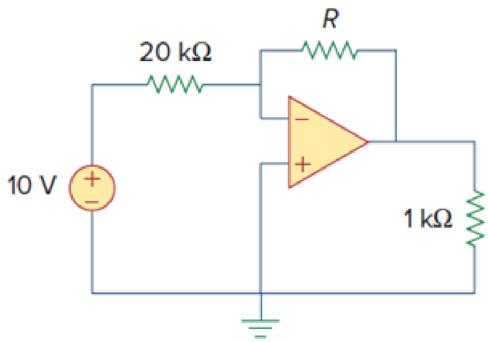
Chapter 5, Problem 85P

Problem

In the op amp circuit of Fig. 5.104, find the value of *R* so that the power absorbed by the 10-kΩ resistor is 10 mW. Determine the power gain.

Figure. 5.104:



Step-by-step solution

Step 1 of 5

Consider the following circuit:

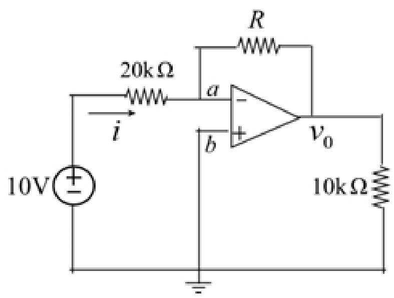


Figure 1

Step 2 of 5

Consider that the op-amp shown in figure 1 is an inverting amplifier.

Therefore, the output of the op-amp is

$$v_0 = -\left(\frac{R_f}{R}\right)v_i$$

Here, $R_f = R\ \Omega$, $R = 20\,\text{k}\Omega$ and $v_i = 10\,\text{V}$.

Substitute the values.

$$v_0 = -\left(\frac{R}{20\times10^3}\right)10$$

$$v_0 = \left(-\frac{R}{2000}\right)$$

Step 3 of 5

Therefore, the power absorbed by $10\,\text{k}\Omega$ resistor is,

$$P_0 = \frac{v_o^{\,2}}{R_L}$$

Here, $v_0 = -\frac{R}{2000}$, $R_L = 10\,\text{k}\Omega$ and $P_0 = 10\,\text{mW}$.

Substitute the values.

$$10\times10^{-3} = \frac{\left(-\frac{R}{2000}\right)^2}{10\times10^3}$$

$$\left(-\frac{R}{2000}\right)^2 = 100$$

$$R = 20\,\text{k}\Omega$$

Therefore, the value of R is $\boxed{20\,\text{k}\Omega}$.

Step 4 of 5

Consider that $v_a = v_b = 0\,\text{V}$ for an ideal op-amp. Since, no current flows into the input terminals of the op-amp.

Hence, the input current is

$$i = \frac{10 - v_a}{20\,\text{k}\Omega}$$

Substitute 0 for v_a.

$$\begin{aligned} i &= \frac{10-0}{20\,\text{k}\Omega} \\ &= 0.5\,\text{mA} \end{aligned}$$

Therefore, the input power is,

$$\begin{aligned} P_{in} &= v_i i \\ &= 10\times0.5\times10^{-3} \\ &= 5\,\text{mW} \end{aligned}$$

Step 5 of 5

Therefore, the power gain is

$$\begin{aligned} P_{gain} &= \frac{P_o}{P_{in}} \\ &= \frac{10\,\text{mW}}{5\,\text{mW}} \\ &= 2 \end{aligned}$$

Therefore, the power gain is $\boxed{2}$.